

Supplementary Material: Liu et al. Reversibility controls on extreme methane clumped isotope signatures from anaerobic oxidation of methane

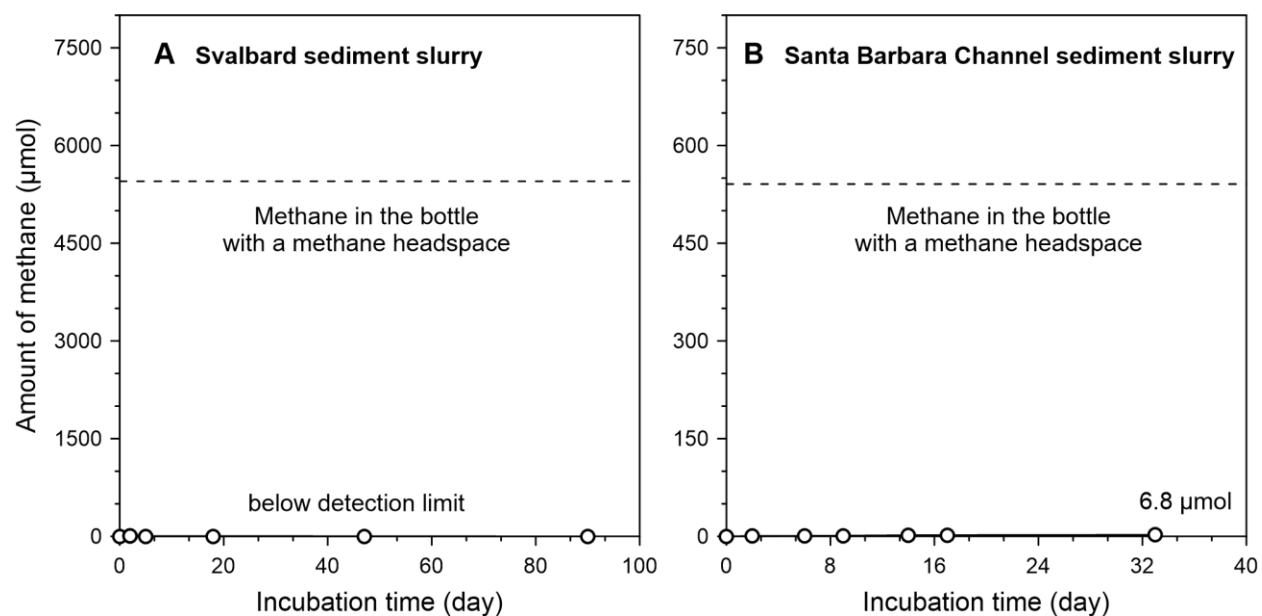


Fig. S1Methane production (amount of methane over time) in the methane-free control experiment: (A) Svalbard sediment slurry, (B) Santa Barbara Channel sediment slurry. For reference, dashed lines represent the amount of methane in the bottle with a pure methane headspace. The control incubation of Santa Barbara slurry showed a small amount of methane build-up in the headspace of the sediment slurry over the 33 days. We interpret this methane increase as the result of equilibration between the slurry and gaseous phases rather than microbial methanogenesis, because unlike the Svalbard sediment slurry (flushed three times), the Santa Barbara Channel sediment slurry was flushed only one time following previous experiments with a pure methane headspace. However, if the methane build-up was the result of microbial methanogenesis, it would represent a very small fraction (ca. 1%) of the methane that is normally added during our experiments, resulting in a negligible isotope effect.

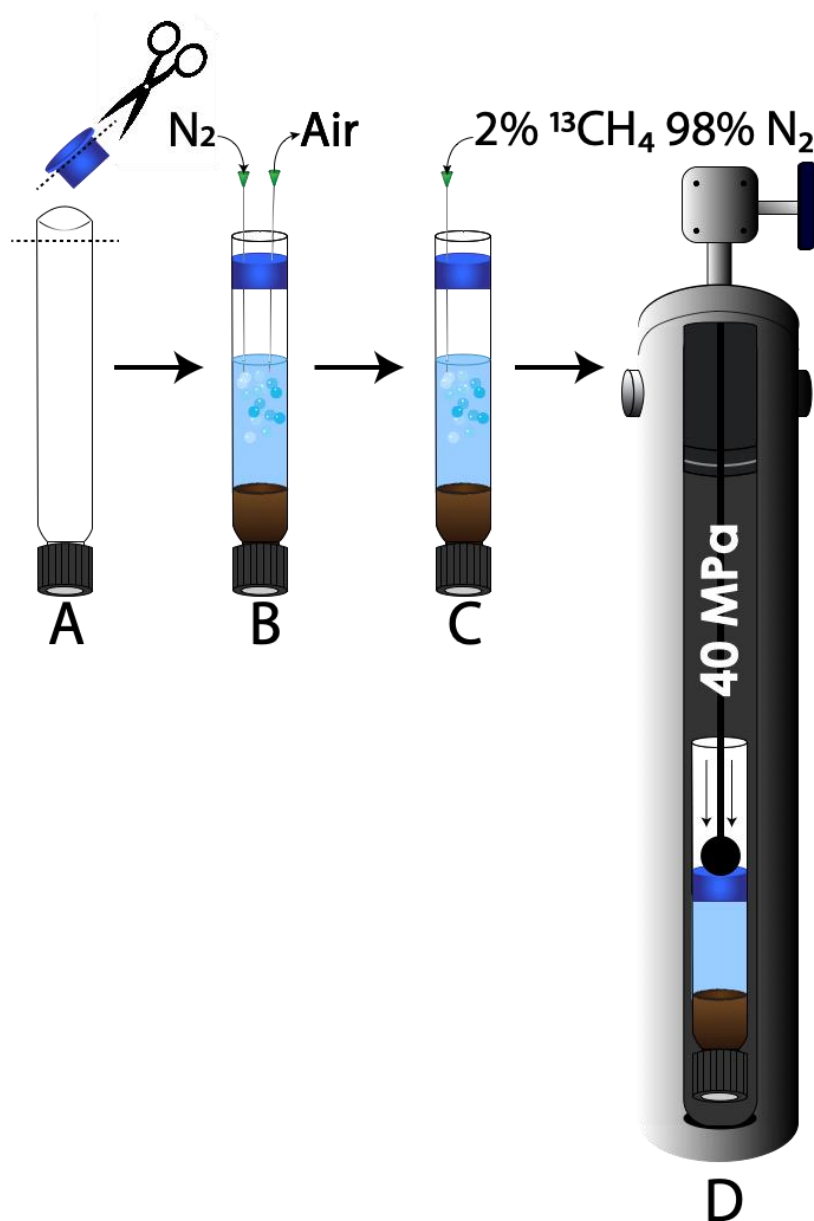


Fig. S2 High pressure cultivation scheme. (A) The bottoms of Hungate tubes are removed using a glass cutter and replaced with a hand-made piston prepared from a butyl rubber stopper. (B) Headspace is sparged with N_2 gas to remove lab air following anaerobic addition of sediment and media. (C) Headspace is replaced with N_2 atmosphere with ^{13}C -labeled CH_4 . (D) Cross-section of stainless-steel hydrostatic pressure vessel containing inverted Hungate tubes. Pressurization to 40 MPa pushes the $^{13}\text{CH}_4$ headspace into solution by way of the butyl rubber piston.

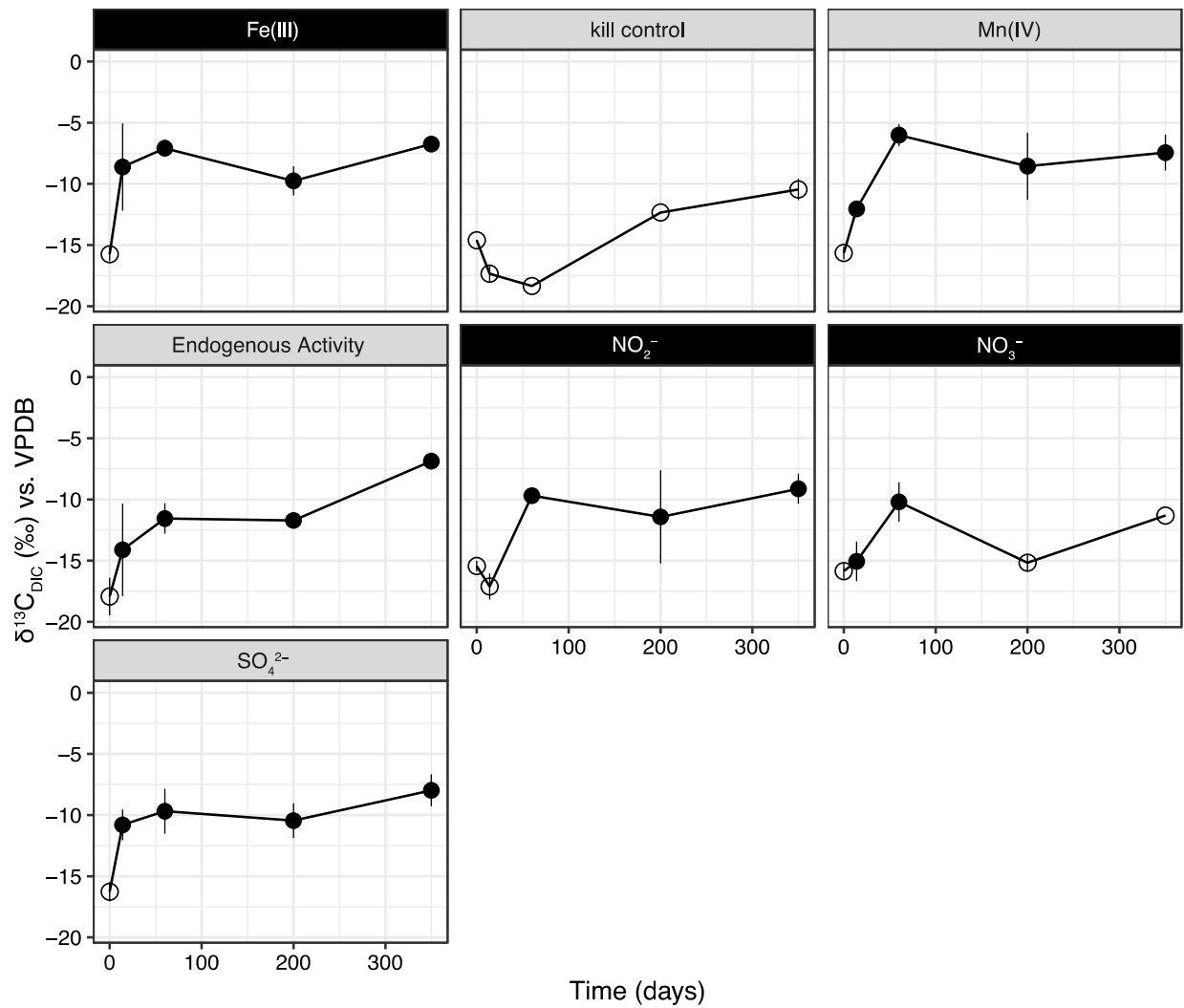


Fig. S3 $\delta^{13}\text{C}_{\text{DIC}}$ in high pressure (40 MPa) Nankai deep biosphere sediments incubated with $^{13}\text{CH}_4$. Rates were calculated relative to an autoclaved kill control according to Eq. S1. Black facets indicate electron acceptor conditions where $\Delta^{13}\text{CH}_3\text{D}$ and $\Delta^{12}\text{CH}_2\text{D}_2$ were measured in parallel experiments incubated at ambient pressure (ca. 150 kPa) and with 100% natural abundance methane. Significance (closed circles) was determined at each time point if $\delta^{13}\text{C}_{\text{DIC_sample}} > (\delta^{13}\text{C}_{\text{DIC_kill}} + 3 \cdot \text{SD}_{\text{kill}})$. $\delta^{13}\text{C}_{\text{DIC}}$ was calculated according to Eq. S1.

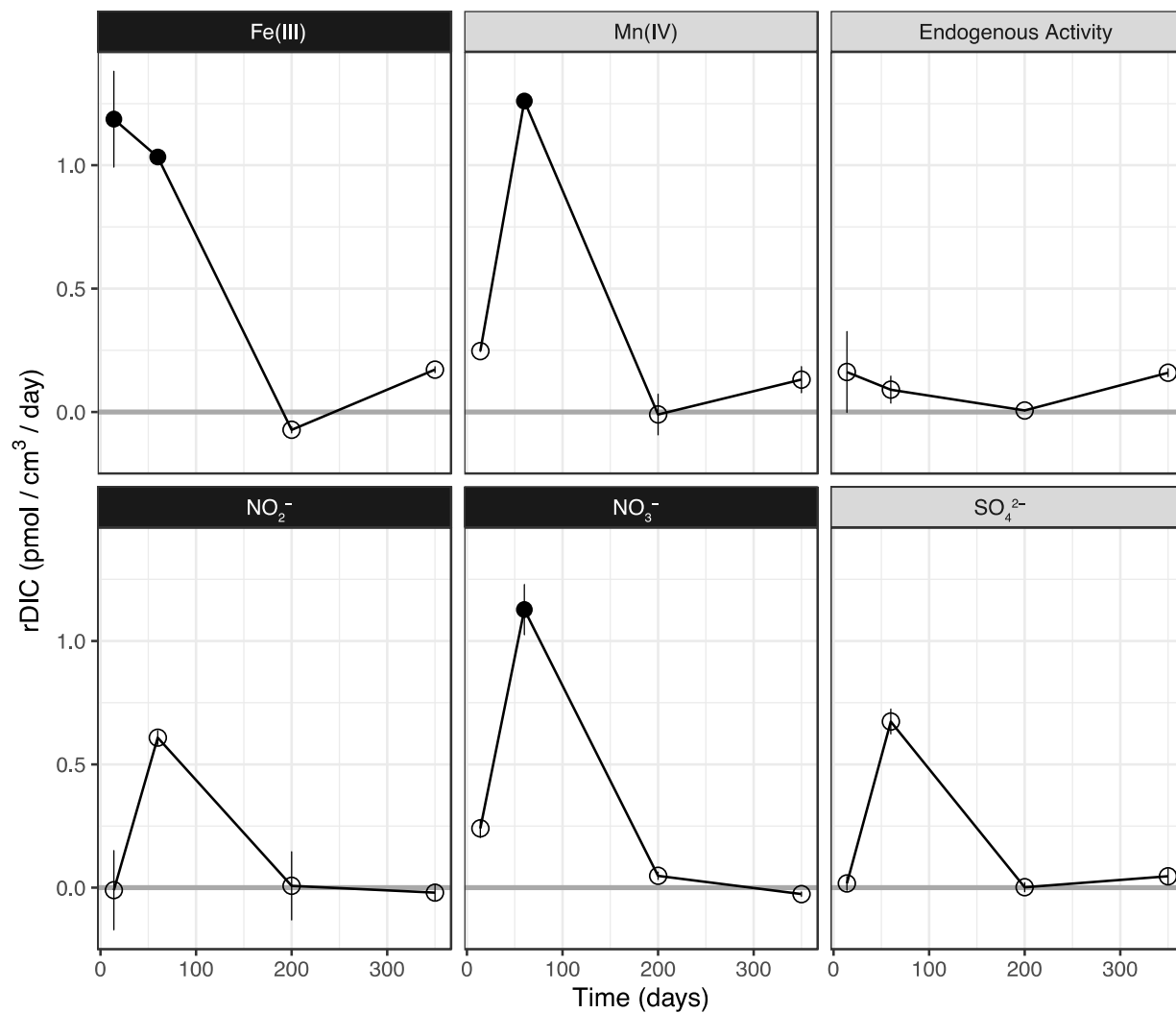


Fig. S4 Rates of AOM $^{13}\text{C}_{\text{DIC}}$ production (r_{DIC}) in high pressure (40 MPa) Nankai deep biosphere sediments incubated with $^{13}\text{CH}_4$. Rates were calculated relative to an autoclaved kill control (normalized here as $r_{\text{DIC}} = 0$) according to Eq. S2. Black facets indicate electron acceptor conditions where $\Delta^{12}\text{CH}_2\text{D}_2$ and $\Delta^{13}\text{CH}_3\text{D}$ were measured in parallel experiments incubated at ambient pressure (ca. 150 kPa) and with 100% natural abundance methane. Closed circles indicate statistically significant r_{DIC} relative to the kill control (one-tailed test, $P < 0.05$).

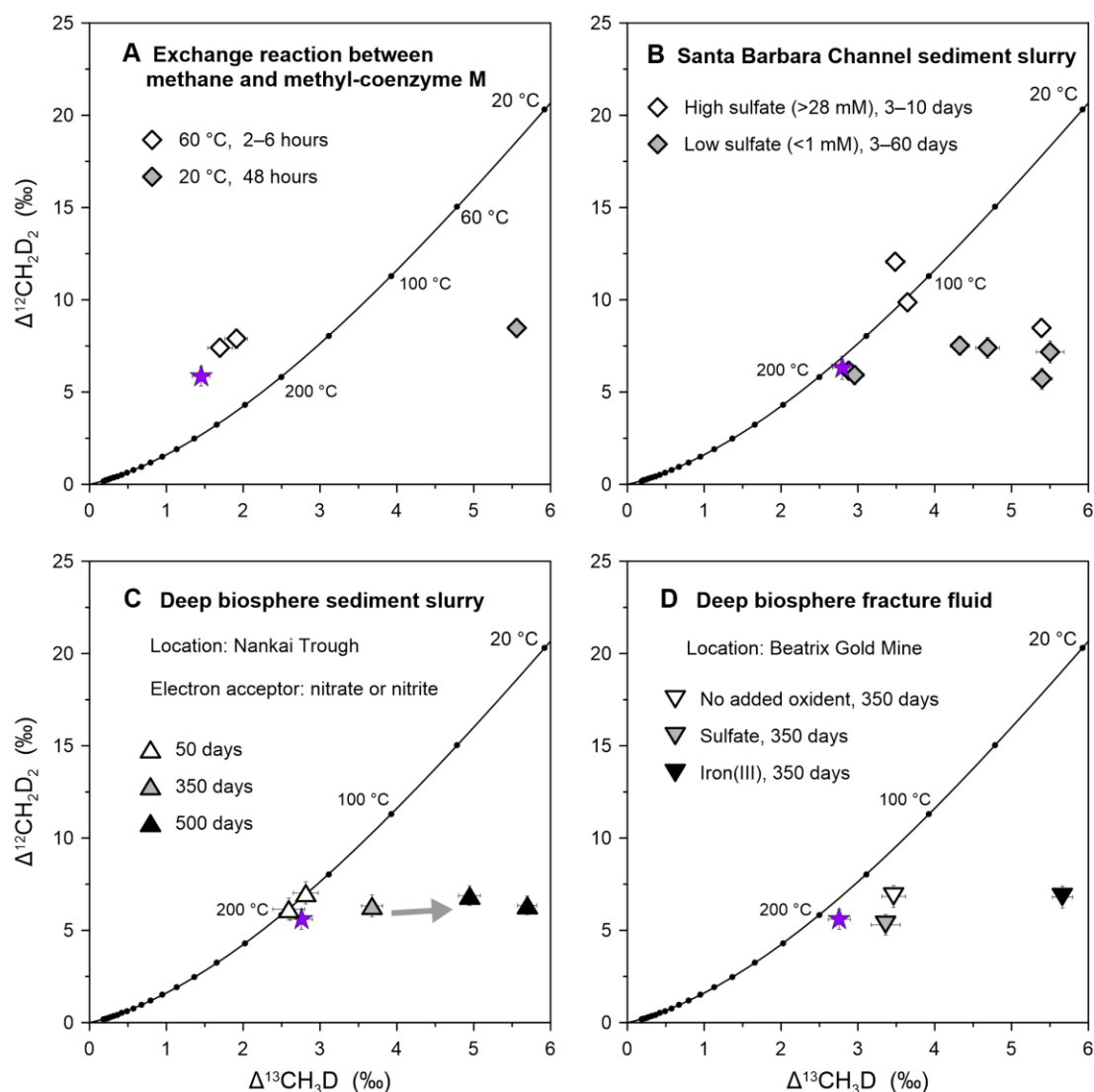


Fig. S5 Equilibrium-driven methane isotopologue data with detailed description for residual methane in incubation experiments. (A) Exchange reaction between methane and methyl-coenzyme M. (B) Santa Barbara Channel sediment incubation with sulfate. (C) Nankai Trough sediment incubation with nitrate or nitrite. (D) Beatrix Gold Mine fracture fluid incubation with sulfate or iron (III). Electron acceptor, incubation time and temperature are shown in the legend of each panel. The purple stars represent the initial tank gas. The solid black line depicts theoretical thermodynamic equilibrium abundances of methane isotopologues, along with corresponding temperatures. Error bars are one standard error.

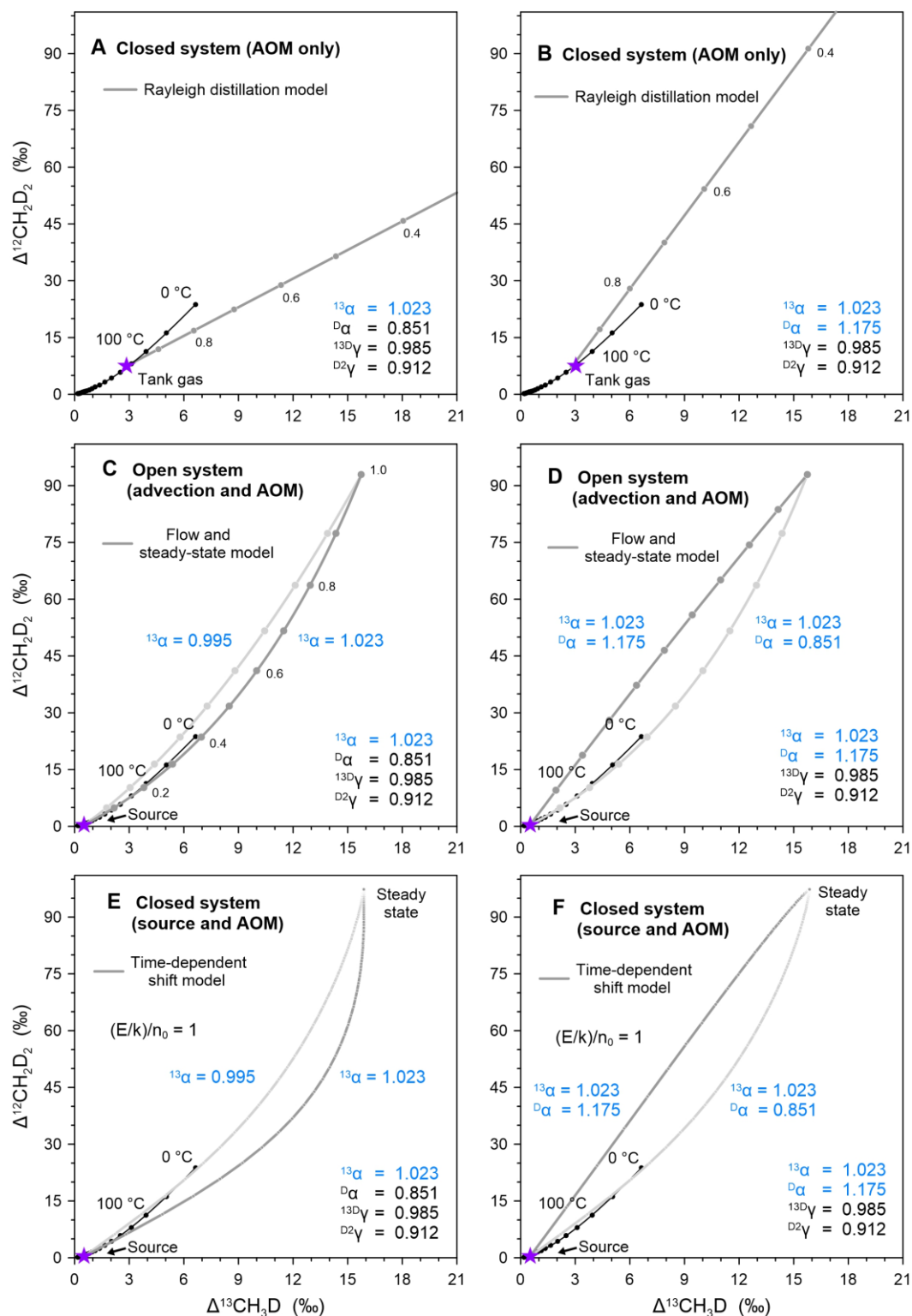


Fig. S6 Sensitivity test of the Rayleigh fractionation model (A–B), open-system flow and steady-state model (C–D) and closed-system time-dependent shift model (E–F). Bulk isotope

fractionation factors in blue are changed from the original values shown in Fig. 6. The carbon isotope fractionation factor (1.023) is adopted from the low sulfate experiment of Wegener et al. (2021). A higher hydrogen isotope fractionation factor of 1.175 is used here as a maximum estimation. Lines and markers follow Fig. 6.

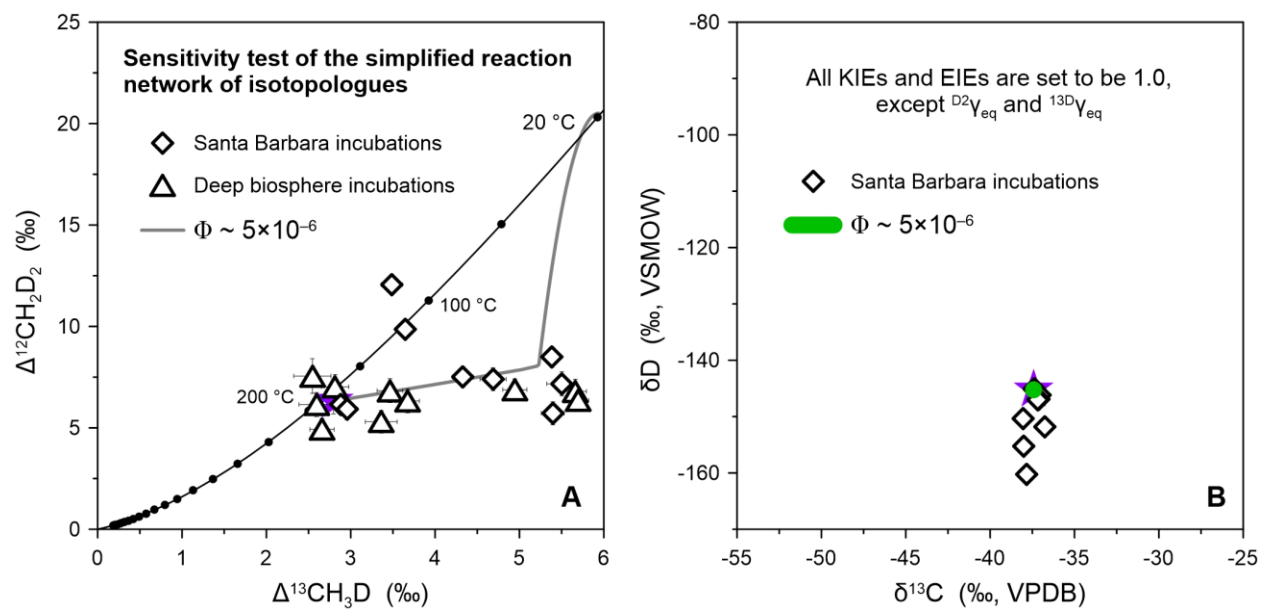


Fig. S7 Sensitivity test of the general model for methane isotopologue fractionation due to AOM. Instead of using the fractionation factors in Table 2, all KIEs and EIEs are set to 1.0 with the exception of $^{12}\text{D}_2\gamma_{\text{eq}}$ and $^{13}\text{D}\gamma_{\text{eq}}$. Lines and markers follow Figs. 7 and 8.

SUPPLEMENTARY METHODS

¹³CH₄ tracer microcosm experiment at high hydrostatic pressure – Nankai Trough

Microcosms were set up inside an anaerobic glove bag (Coy Laboratory Products, Grass Lake, MI USA) under a 5:95 H₂:N₂ atmosphere. Fresh aluminum foil was placed on the working surface of the glove bag and ethanol-sterilized nitrile gloves were worn over the glove bag's neoprene gloves to minimize potential contamination of low-biomass samples. The surface of each C0023A whole round core (WRC) was scraped using an ethanol-sterilized scalpel to remove sediment exposed to Kochi Core Center's anaerobic packaging. For each WCR, ~ 1 g (wet weight) of freshly exhumed interior sediment was transferred into high-pressure modified Hungate tubes (Fig. S2; Bowles et al., 2011) containing 5 mL anoxic sulfate-free artificial seawater medium at pH 8.0 (Widdel and Bak, 1992; Holler et al., 2011) and supplemented with 10 mM (final concentration) of one of the following electron acceptors: SO₄²⁻, NO₂⁻, NO₃⁻, Mn⁴⁺ (in the form of MnO₂), or Fe³⁺ (in the form of Fe₂O₃•H₂O), in addition to a no-added oxidant control (i.e., “endogenous activity”). Fe₂O₃•H₂O and MnO₂ were synthesized as previously described (Cornell and Schwertmann, 2003; Händel et al., 2013). The headspace of each microcosm was replaced with 2:98 ¹³CH₄:N₂ (100 kPa), prepared by mixing via gas-tight syringes (Trajan Scientific and Medical, Melbourne, Australia) in sealed Tedlar[®] bags (DuPont, Wilmington, DE USA). An autoclaved kill control was included as a reference for determining significant biological AOM. All treatments were assessed in biological triplicate.

Microcosms were inverted and placed inside stainless-steel hydrostatic vessels with a pin-retained piston closure pre-heated in gravity ovens to an approximate *in-situ* temperature of 40°C (Fig. S2). Vessels were sealed and pressurized to 40 ± 0.5 MPa using a hydrostatic pump as previously described (Yayanos, 1995) and subsequently returned to gravity ovens for incubation.

Subsamples of the enrichments were collected at 0, 14, 60, 200, and 350 days to monitor CH₄ concentrations and the evolution of $\delta^{13}\text{C}_{\text{DIC}}$. Briefly, hydrostatic vessels were removed from ovens and depressurized for retrieval of Hungate tube microcosms. From each sample, a 500 μL aliquot of the slurry was anaerobically transferred using Ar-sparged syringes and needles into combusted borosilicate serum vials pre-treated with saturated HgCl₂ to kill any microbes contained in the sample (Sherwood Lollar et al., 1993a; Sherwood Lollar et al., 1993b). Vials were sealed under a 100% N₂ atmosphere with crimped 0.1 NaOH-treated butyl rubber stoppers (Bellco Glass, Inc., Vineland, NJ USA). Following sample transfer, serum vials were over-pressurized with N₂ gas to minimize the risk of mixing with atmospheric CO₂. To exsolve all DIC out of solution, vials were supplemented with 100 μL of 0.5 N H₃PO₄, and heated overnight in a water bath at 70°C. CH₄ concentrations and $\delta^{13}\text{C}_{\text{DIC}}$ were analyzed from the headspace using a Picarro cavity ringdown spectrometer equipped with a G2101-I Isotopic CO₂ analyzer (Picarro, Inc., Sunnyvale, CA USA). Resulting isotopic values were calculated according to

$$\delta^{13}\text{C}_{\text{DIC}} = \left(\frac{\left(\frac{^{13}\text{C}_{\text{DIC}}}{^{12}\text{C}_{\text{DIC}}} \right)_{\text{sample}}}{\left(\frac{^{13}\text{C}_{\text{DIC}}}{^{12}\text{C}_{\text{DIC}}} \right)_{\text{standard}}} - 1 \right) \times 1000 \text{ ‰} \quad (\text{S1})$$

with respect to the [¹³C/¹²C] ratio of the Vienna Pee Dee Belemnite standard. AOM was determined to have a significant influence on $\delta^{13}\text{C}_{\text{DIC}}$ if its average $\delta^{13}\text{C}_{\text{DIC}} \pm$ standard deviation (SD) was greater than the average $\delta^{13}\text{C}_{\text{DIC}} + 3 \times \text{SD}$ of the kill control ($\delta^{13}\text{C}_{\text{DIC},d.\text{kill}}$) (Fig. S3).

DIC production rates (r_{DIC}) for each sample are shown in Figure S4 and were calculated according to

$$r_{\text{DIC}} \text{ (pmol cm}^{-3} \text{ day}^{-1}) = \frac{\left(\frac{\delta^{13}\text{C}_{\text{DIC},d} - \delta^{13}\text{C}_{\text{DIC},d.\text{kill}}}{\delta^{13}\text{C}_{^{13}\text{CH}_4} - \delta^{13}\text{C}_{\text{DIC},d.\text{kill}}} \right)_{\text{DIC},d} \left(\frac{\rho_d}{\rho_g} \right)}{t} \quad (\text{S2})$$

where DIC_d is the sum of $^{13}C_{DIC}$ and $^{12}C_{DIC}$ in pmol at depth d , ρ_d is the bulk density of the sediment at depth d in grams cm^{-3} (<http://sio7.jamstec.go.jp/j-cores.data/370/C0023A/>), and g_d is the grams of sediment incubated from depth d , and t is the number of days of incubation. Data were assumed to follow a normal distribution and r_{DIC} values were assigned z-scores according to

$$z = (r_{DIC_sample} - \mu_{kill}) / \sigma_{kill} \quad (S3)$$

from which P values were subsequently calculated using a 1-tailed test.

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